

IV. RESULTS AND ANALYSIS

A. Exploratory Data Analysis (EDA) Findings

Prior to modeling, a 1% sample of the 24-million-record dataset was analyzed. Key observations included:

- **Feature Correlation:** A strong linear correlation was observed between `trip_distance` and `fare_amount`. However, `trip_duration` exhibited a high degree of variance relative to distance, suggesting that external factors like traffic density play a critical role.
- **Data Quality:** The application of the 3-Sigma rule effectively removed non-physical outliers, resulting in a cleaner distribution for gradient-based learning.
- **Noise Reduction:** Though EDA's data cleaning, the number of rows was reduced from over 24 million to 17.69 million. Over a quarter of the data recorded was invalid, majority with some zero/negative values.

B. Duration Prediction (Hypothesis 1)

The comparison between Linear Regression and Random Forest (RF) for trip duration highlighted the limitations of linear assumptions in urban traffic modeling.

TABLE II
DURATION PREDICTION PERFORMANCE

Model	RMSE	MAE	R^2
Linear Regression	7.33	4.86	0.690
Random Forest	5.70	3.60	0.812

The RF model achieved an R^2 of **0.812**, a significant improvement over the 0.690 achieved by the baseline. Analysis of RF feature importance revealed that while `trip_distance` (0.678) is the primary driver, `PULocationID_index` (0.251) and `pickup_hour` (0.053) are vital for capturing non-linear congestion patterns.

C. Fare Prediction (Hypothesis 2)

Fare prediction yielded the highest accuracy across all regression tasks. This could be expected due to the regulated pricing structure of NYC taxi fares, where cost is primarily determined by distance and predefined tariff rules. [2]

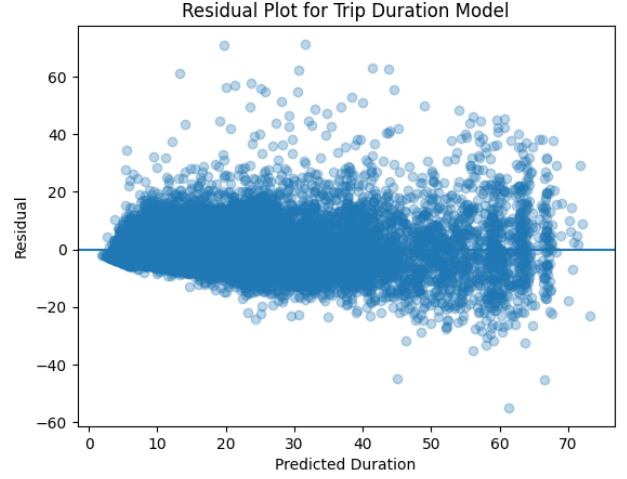


Fig. 3. Residual Plot for Trip Duration on 1% prediction sample

TABLE III
FARE PREDICTION PERFORMANCE

Model	RMSE	MAE	R^2
Linear Regression	5.89	2.89	0.887
Random Forest	5.55	2.52	0.900

The feature importance analysis revealed a notable insight: `rush_hour` (0.281) was significantly more predictive than `trip_duration` (0.003). Since `fare_amount` represents the base metered price (excluding surcharges), this suggests cost of a trip is more strongly correlated with predefined peak traffic windows than with the raw elapsed time of the trip. Effectively, the `rush_hour` feature acts as a proxy for traffic congestion levels, proving more statistically significant than the total duration itself.

D. Trip Classification (Hypothesis 3)

The binary classification task (Short vs. Long trips) was evaluated using ROC-AUC and F1-score to account for the class distribution.

TABLE IV
TRIP CLASSIFICATION PERFORMANCE

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.860	0.857	0.931
Random Forest	0.878	0.877	0.942

The RF Classifier achieved a near-perfect ROC-AUC of **0.942**. Feature importance for classification was dominated by `trip_distance` (0.838), fol-